

EcoSort AI

Problem Statement

Improper waste segregation is a major challenge for effective recycling and sustainable waste management. Many people do not know whether a waste item belongs to organic, recyclable, hazardous, or e-waste categories. As a result, recyclable materials end up in landfills, hazardous waste contaminates soil and groundwater, e-waste is improperly discarded, and organic waste produces harmful methane emissions instead of being composted.

Project Overview

Our team has developed an AI-powered waste classification system that uses computer vision and deep learning to identify waste from uploaded images and guide users toward environmentally responsible disposal. The system uses MobileNetV2 transfer learning to classify waste into four categories: organic, recyclable, hazardous, and e-waste. It also provides disposal recommendations, calculates environmental impact, tracks sustainability scores, and displays waste analytics through an interactive dashboard.

Solution Offered

The EcoSort AI system consists of:

- An AI-powered image classification model that identifies organic, recyclable, hazardous, and e-waste
- A recommendation engine that explains how to correctly dispose of, reuse, or recycle each identified waste item.
- A carbon impact calculator that estimates CO₂ and energy savings from proper waste disposal.
- A sustainability scoring system that rewards users for responsible waste classification and disposal.

Who Are The End Users?

- Individuals and households seeking guidance on correct waste disposal.
- Schools, colleges, and educational institutions teaching sustainability and environmental awareness.
- Municipal and community waste management programs seeking low-cost citizen education tools.
- Environmental NGOs conducting recycling and sustainability awareness campaigns.

Technology Used To Solve The Problem

Artificial Intelligence & Computer Vision

- Python: Main programming language used to develop the complete system.
- TensorFlow / Keras: Used to build, train, and evaluate the deep-learning classification model.
- OpenCV: Used for image processing, resizing, and color-space conversion.

Data Handling, Analytics & Deployment

- NumPy & Pandas: Used for numerical computations, image processing, CSV-based activity logging.
- Matplotlib & Scikit-learn , Google Colab, ipywidgets & GitHub